Multiple Choice Section

Q5

A rock suspended by a weighing scale weighs 25 N out of water, and 10 N when submerged in water.
What is the buoyant force on the rock?

A. 15 N
B. 10 N
C. 40 N
D. none of these

Q6

Light bends in a gravitational field. Why isn't this bending taken into consideration by surveyors who use laser beams as straight lines?

A. The bending of light occurs only in outer space.
B. Light travels so fast that the curvature is not noticeable for surveying purposes.
C. Laser light is an exception to this rule.
D. Red light doesn't bend so much, which is why lasers are red.

Q7

Scientific evidence supports the date of the Big Bang event at

A. 7000 years ago
B. 5 billion years ago
C. 100,000 years ago
D. about 14 billion years ago

Q8

The star Polaris is always directly over

A. the equator
B. any location north of the equator

C. the North Pole
D. the South Pole

Q9

If you toss a ball straight upward at 30 m/s with no air resistance, 6 seconds later its speed is

A. 0 m/s
B. 30 m/s
C. 20 m/s
D. 10 m/s

Q10

A 50 N object falling in air experiences 30 N of air resistance.
The amount of net force on the falling object is

A. 20 N
B. 50 N
C. 80 N
D. 0 N

Q11

A girl pushes with a 120 N horizontal force on a 20 kg crate of coffee resting on a warehouse floor.
The force of friction on the crate is 60 N.
What is the acceleration of the crate?

A. 0 m/s^2
B. 9 m/s^2
C. 6 m/s^2
D. 3 m/s^2

Q12

A 10 kg bag of groceries is tossed onto a table at a speed of 5 m/s and slides to a stop in 2 s. The direction of the friction force acting on the bag is

A. The same as the direction of motion
B. Opposite to the direction of motion
C. Downward
D. Upward

Q13

If your mass remained the same while the mass of Earth doubled while staying the same size, your weight would

A. be the same
B. double
C. decrease to one-half
D. quadruple

Q14

What is the amount of heat Q needed to warm 100 kg of water from 20°C to 37°C?
Given: c = 4.18 kJ/(kg·°C)

A. 3550 kJ
B. 7730 kJ
C. 7100 kJ
D. 17000 kJ

Q15

You heat a half-cup of tea and its temperature rises by 16°C.
How much will the temperature rise if you add the same amount of heat to a full cup of tea?

A. 16°C
B. 8°C
C. 4°C
D. 2°C

Q16

The quantity of heat needed to raise the temperature of a certain substance a specific amount is 1 Calorie (with capital C).
This is the same amount of energy as

A. 1000 calories
B. 4180 calories
C. Neither of these
D. Both of these

Q17

A bimetallic strip used in thermostats relies on the fact that different metals have different

A. rates of thermal expansion
B. thermal energies at different temperatures
C. specific heat capacities
D. all of these

Q18

A power line with a resistance of 4 Ω draws a current of 40 A.
The power dissipated in the line is

A. 320 W
B. 160 W
C. 640 W
D. 6400 W

Q19

The nucleus of an atom is charged

A. positively
B. negatively
C. both
D. none of the above

Q20

Two charges separated by 1 m exert a 10 N force on each other.
If the charges are pushed to 0.25 m separation, the force on each charge will be

A. 2 N
B. 16 N
C. 5 N
D. 4 N

Q21

If a potential difference of 24.0 V is required to produce a current of 3.0 A in a wire, the resistance of the wire is

A. 8.0 Ω
B. 27 Ω
C. 0.50 Ω
D. 6.0 Ω

Writing Section

Problem 1

A rock is thrown horizontally from a bridge and hits the water below.
The rock travels a smooth parabolic path and is airborne for t = 2 s.

What is the height of the bridge if the time the rock is airborne is 2 s?

Problem 2

a) How much thermal energy is needed to heat 650 g of copper from 25°C to 75°C?
Given: specific heat of copper = 390 J/(kg·°C)

b) A block of ice at 0°C is dropped and completely melts upon impact.
Assume no air resistance and that all energy goes into melting the ice.
Show that the height required is at least 34 km.
Given: latent heat of fusion of water = 335,000 J/kg

Problem 3

a) How many kilometers is a light-year?
Given: speed of light $c = 300 \times 10^3$ km/s

b) In the film *Avatar*, humans travel from Earth to Pandora, about 4.3 light-years away.
If the spacecraft travels at 1200 km/h, how long will the journey take?
Express your answer in years.

c) If the spacecraft can travel at 0.2c (20% of the speed of light), how long will the journey take?
Express your answer in years.